RESEARCH ARTICLE

Autoimmune Diseases and Incident Spontaneous Subarachnoid Hemorrhage

A Swedish Population-Based Cohort Study

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Neurology® 2026;106:e214679. doi:10.1212/WNL.0000000000214679

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Abstract

Background and Objectives

Compelling evidence documents an association between autoimmune diseases and several types of cardiovascular diseases. Knowledge on whether autoimmune diseases may increase the risk of subarachnoid hemorrhage (SAH), a rare but severe type of cerebrovascular event, is very limited. The aim of this study was to determine the association between autoimmune diseases and SAH.

Methods

We conducted a nationwide cohort study including individuals who were parents of all live births recorded in the Swedish Medical Birth Register during 1973–2014 and who were alive, resided in Sweden, and had no history of SAH at study baseline. We obtained data on autoimmune diseases, SAH, and covariates through linkage to several population-based registers and followed participants from January 1, 2001, until SAH diagnosis, death, emigration, or December 31, 2023. We used Cox proportional hazards models to calculate hazard ratios (HRs) and 95% confidence intervals (CIs) for SAH in relation to autoimmune diseases.

Results

Among the 3,824,528 study participants (mean age at baseline 38.1 (±13.2) years, 55.2% female), 608,043 (15.9%) had a record of any autoimmune disease before or during the study period; 10,311 (0.3%) were diagnosed with SAH over the follow-up period of 83.5 million person-years. Overall, any autoimmune disease was associated with an increased risk of SAH (HR = 1.18; 95% CI 1.11–1.25). Several categories, as well as some specific autoimmune diseases, that is, type 1 diabetes (HR = 1.36; 95% CI 1.15–1.61), rheumatoid arthritis (HR = 1.29; 95% CI 1.10–1.50), systemic lupus erythematosus (HR = 1.49; 95% CI 1.01–2.21), psoriasis vulgaris (HR = 1.18; 95% CI 1.04–1.33), multiple sclerosis (HR = 1.35; 95% CI 1.02–1.78), primary biliary cirrhosis (HR = 2.66; 95% CI 1.57–4.49), and IgA nephropathy (HR = 1.42; 95% CI 1.08–1.87), were associated with an increased risk of SAH.

Discussion

This study suggested that a broad range of autoimmune diseases may be involved in the etiology of SAH. If confirmed by future research, the results may inform the development of targeted SAH prevention strategies for individuals with autoimmune diseases; however, reliance on register-based diagnoses and the lack of detailed lifestyle information such as smoking may have led to misclassification and residual confounding.

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Glossary

CVD = cardiovascular disease; **DM** = diabetes mellitus; **FDR** = false discovery rate; **HR** = hazard ratio; **ICD** = International Classification of Diseases; **MBR** = Medical Birth Register; **PR** = Patient Register; **SAH** = subarachnoid hemorrhage; **SLE** = systemic lupus erythematosus.

Introduction

Spontaneous subarachnoid hemorrhage (SAH), most commonly caused by rupture of intracranial aneurysms, is an uncommon yet devastating form of stroke.[1] Globally, the incidence of SAH is estimated at approximately 8.3 per 100,000 individuals in 2021.[2] Although age-standardized burden rates of SAH have decreased, the absolute number of cases has continued to rise. Subarachnoid hemorrhage remains among the leading cardiovascular and neurologic causes of death and disability worldwide.[2] While risk factors such as hypertension, alcohol abuse, smoking, and familial predisposition have been established, the underlying pathophysiology of aneurysm formation and rupture is not fully understood.[2,3] Emerging evidence suggests that chronic vascular inflammation and immune-mediated processes may play a role in weakening the arterial wall, contributing to aneurysm instability and subsequent hemorrhage.[3-6]

Autoimmune diseases encompass a diverse group of disorders characterized by dysregulated immune responses targeting the body's own tissues. In recent years, growing evidence has highlighted the role of systemic inflammation and immune dysregulation—key features of autoimmune diseases—in the development and progression of cardiovascular and cerebrovascular diseases.[7,8] Autoimmune diseases have been associated with increased risks of various vascular events, including stroke,[9-12] myocardial infarction,[13,14] and venous thromboembolism,[15] and chronic inflammation, endothelial dysfunction, and autoantibody-mediated vascular injury have been proposed as central mechanisms. Several studies have explored the relationship between autoimmune diseases and stroke and suggested an increased stroke risk particularly among individuals with conditions such as systemic lupus erythematosus (SLE),[16-19] rheumatoid arthritis,[20-22] and vasculitis.[23] However, most existing research has focused on ischemic rather than hemorrhagic stroke, and data specifically linking autoimmune diseases to SAH are scarce and mainly focused on SLE.[16-19]

The aim of this study was to address this gap by systematically investigating the association between autoimmune diseases and incident spontaneous SAH in a large, population-based cohort from Sweden.

Methods

Study Population

We conducted a cohort study by linking data from 8 Swedish registers with national coverage. This study was based on an existing parental cohort originally established by linking the Medical Birth Register (MBR) to the Multi-Generation Register. A detailed description of the cohort, the linked registers, and the retrieved variables has been provided in an earlier article.[24] Shortly, the study population included the parents of the 4,298,283 live births recorded in the MBR from 1973 to 2014. Parents were identified through the Multi-Generation Register; information on virtually all mothers and on 83% of fathers was available to us. Participation in our study further required valid information on sex and year of birth, as well as the possibility of linkage across national registers through the unique personal identity number. Individuals with missing or inconsistent linkage information were excluded. The resulting cohort of 3,924,239 parents was further linked to the Patient Register (PR), the Cause of Death Register, the Total Population Register, the Longitudinal Integrated Database for Health Insurance and Labour Market Studies, the Education Register, and the Register of Incomes and Taxes to collect relevant data on exposures, outcomes, and covariates. This study included individuals who were alive, resided in Sweden, and had no history of SAH at study baseline (defined using the International Classification of Diseases (ICD) codes in eTable 1). We defined baseline as January 1, 2001, or date of immigration to Sweden after January 1, 2001, whichever occurred later. The reason for choosing 2001 for the definition of baseline was that the coverage of the specialized outpatient care in PR—relevant for identification of autoimmune diseases—became nationwide in 2001.

Measures

Exposures

The primary exposure in this study was overall autoimmune disease, identified from the PR using the ICD codes presented in eTable 1; the register was established in 1964 and attained nationwide coverage for all inpatient care in 1987 and for nonprivate specialized outpatient care in 2001. Analyses were also conducted for 9 categories of autoimmune diseases as separate exposures, that is, endocrine autoimmune diseases, inflammatory arthritis, vasculitis, connective tissue autoimmune diseases, skin autoimmune diseases, blood autoimmune diseases, nervous system autoimmune diseases, gastrointestinal autoimmune diseases, and other autoimmune diseases, following a previously published classification.[25] We searched the PR data for both primary and secondary diagnoses to identify individuals with autoimmune diseases. In case of several autoimmune diseases, we considered the individual's earliest diagnosis. We considered exposures as time-varying; that is, individuals contributed to the unexposed person-time

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from baseline until the date of the first autoimmune disease diagnosis (if there was such a diagnosis) and to the exposed person-time afterward. Individuals with a record of autoimmune diseases before baseline contributed to the exposed group throughout the study.

Outcome

We identified nontraumatic SAH cases by searching the main diagnosis in the inpatient component of the PR and the underlying causes of death in the Cause of Death Register using the ICD codes listed in eTable 1. Follow-up ended at the earliest of the following events: SAH diagnosis, emigration (available until December 31, 2014), death, or December 31, 2023. Because the study baseline was defined as January 1, 2001 (or the date of immigration thereafter), individuals with any history of SAH before baseline were excluded. Consequently, all SAH events that occurred during follow-up—and were included in the analyses—were identified using the ICD-10 code I60, which specifically denotes ruptured aneurysmal SAH.

Covariates

Eight covariates were retrieved for the analysis, including age at baseline, sex, marital status, country of birth, household income (later categorized based on the tertile distribution of each 10-year interval), highest educational attainment, history of any cardiovascular diseases (CVDs), and history of diabetes mellitus (DM). Information on birth year, sex, marital status, and country of birth was retrieved from the Total Population Register. Data on income were obtained from the Register on Incomes and Taxes while those on education from the Education Register. Information on history of any CVD and history of DM was retrieved from the PR. Information on marital status, educational level, and household income was collected for the calendar year before study enrollment. If data from that specific year were unavailable, the closest available data within the past 5 years were used.

For sensitivity analyses, we retrieved data on (1) women's smoking, obesity, hypertension, and DM before childbirth from the MBR; (2) smoking and tobacco dependence; and (3) family history of CVDs and autoimmune diseases before baseline, as well as on several medical conditions including DM, hypertension, ischemic heart disease, cerebrovascular disease, atrial fibrillation, heart failure, peripheral artery disease, and valvular heart disease from the PR. We used the ICD codes listed in eTable 1 and categorized variables given in Table 1.

Statistical Analysis

Main Analysis

We ran Cox regression models to estimate the associations between autoimmune diseases and the risk of incident SAH. These models estimated hazard ratios (HRs) and 95% CIs, comparing SAH risk among individuals with autoimmune diseases with those without. Analyses were conducted for 3 major exposure categories: (1) overall autoimmune disease, (2) the 9 categories of autoimmune diseases, and (3) specific types of autoimmune diseases. We ran unadjusted models and models adjusted for sex, age, country of birth, household income, education, marital status, history of DM, and history of CVD. Covariates were modeled with the categories in Table 1, except for age, which was modeled as a categorical variable with 5-year intervals (i.e., 0–19, 20–24, and 25–29 years) to flexibly account for its potentially non–log-linear relationship with SAH. Because Schoenfeld residuals suggested a violation of the proportional hazards assumption, we used flexible parametric survival models to estimate the time-varying effects of autoimmune diseases on the risk of incident SAH.

To study whether the association between autoimmune disease and SAH differed across major sociodemographic groups and according to genetic cardiovascular vulnerability, we performed stratified analyses based on sex (male vs female), age (<50 years vs ≥50 years at baseline), educational attainment (≤9 years, 10–14 years, ≥15 years), household income (low, middle, high), and parental history of CVD.

Sensitivity Analysis

We conducted several sensitivity analyses. First, we further adjusted for (1) a family history of autoimmune disease and CVD in analyses restricted to individuals with register links to their parents; (2) smoking, obesity, DM, and hypertension before childbirth in women; and (3) tobacco use disorder and alcohol use disorder in the whole cohort, in addition to the variables included in the main model. Second, to consider unmeasured confounders shared within families, we conducted sibling analyses for overall autoimmune disease. Third, we repeated the main analyses with an alternative definition of the outcome; that is, we used primary diagnoses from both inpatient and outpatient care and the main cause of death from the Swedish Cause of Death Register to define the outcome. Fourth, we restricted analyses to individuals with only 1 specific autoimmune disease to address the concern that associations may be driven by individuals with multiple autoimmune conditions who might have specific clinical profiles, health care utilization patterns, or comorbidity burdens. Fifth, to examine the role of prevalent and incident autoimmune disease in the studied association and whether the association differs across periods (given that treatments for several autoimmune diseases gradually improved over our study period), we excluded individuals with autoimmune disease diagnosed before baseline and also performed stratified analyses based on whether autoimmune disease was diagnosed before or after baseline and before or after the median follow-up time (July 1, 2012). Sixth, to address concerns that a short-term effect of autoimmune disease on SAH could be influenced by its detection during the diagnostic workup or treatment of a common comorbidity of SAH, we performed analyses by (1) excluding patients with cardiovascular comorbidities recorded within 3 months of their SAH diagnosis, including DM, hypertension, ischemic heart disease, cerebrovascular insult, heart failure, atrial fibrillation, peripheral artery disease, and valvular heart disease; (2) excluding events from the follow-up that occurred during the 3 months after the exposure; and (3) repeating the main

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Table 1 Characteristics of the Study Population by Exposure to Any Autoimmune Disease

	Exposed to autoimmune diseases, n (%)	
Characteristics	**Yes (n = 608,043)**	**No (n = 3,216,485)**
Age (mean ± SD)	41.2 ± 13.3	37.5 ± 13.0
Sex		
Men	227,894 (37.48)	1,486,261 (46.21)
Women	380,149 (62.52)	1,730,224 (53.79)
Country of birth		
Sweden	535,210 (88.02)	2,814,804 (87.51)
Other country	72,833 (11.98)	401,681 (12.49)
Calendar year of baseline		
2001–2005	595,243 (97.89)	3,113,992 (96.81)
2006–2010	8,842 (1.45)	68,172 (2.12)
2011–2014	3,958 (0.65)	34,321 (1.07)
Marital status		
Single, widowed, or divorced	312,187 (51.34)	1,803,971 (56.09)
Married or in registered partnership	294,683 (48.46)	1,401,007 (43.56)
Missing	1,173 (0.19)	11,507 (0.36)
Family income		
Low tertile	174,135 (28.64)	910,397 (28.30)
Middle tertile	248,535 (40.87)	1,308,559 (40.68)
High tertile	180,232 (29.64)	951,756 (29.59)
Missing	5,141 (0.85)	45,773 (1.42)
Highest education		
≤9 y	107,566 (17.69)	467,042 (14.52)
10–14 y	371,066 (61.03)	1,958,833 (60.90)
≥15 y	127,776 (21.01)	777,880 (24.18)
Missing	1,635 (0.27)	12,730 (0.40)
CVD before baseline		
No	555,700 (91.39)	3,089,631 (96.06)
Yes	52,343 (8.61)	126,854 (3.94)
DM before baseline		
No	574,656 (94.51)	3,209,130 (99.77)
Yes	33,387 (5.49)	7,355 (0.23)
Autoimmune diseases before baseline		
No	499,374 (82.13)	3,216,485 (100.00)
Yes	108,669 (17.87)	0 (0.00)
Alcohol use disorder before baseline		
No	571,831 (94.04)	3,068,865 (95.41)
Yes	36,212 (5.96)	147,620 (4.59)

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Table 1 Characteristics of the Study Population by Exposure to Any Autoimmune Disease *(continued)*

Characteristics	Exposed to autoimmune diseases, n (%)	
	Yes (n = 608,043)	No (n = 3,216,485)
Tobacco use disorder before baseline		
No	589,866 (97.01)	3,170,706 (98.58)
Yes	18,177 (2.99)	45,779 (1.42)
Family history of autoimmune diseases		
No	535,319 (88.04)	2,845,435 (88.46)
Yes	11,486 (1.89)	56,241 (1.75)
Missing	61,238 (10.07)	314,809 (9.79)
Family history of CVD		
No	268,815 (44.21)	1,681,752 (52.29)
Yes	277,787 (45.69)	1,218,618 (37.89)
Missing	61,441 (10.10)	316,115 (9.83)
Emigration up to 2014		
No	605,218 (99.54)	3,176,615 (98.76)
Yes	2,825 (0.46)	39,870 (1.24)
Death up to 2023		
No	531,710 (87.45)	2,991,134 (92.99)
Yes	76,333 (12.55)	22,5351 (7.01)
Hypertension before childbirth[a]		
No	366,158 (96.32)	1,675,784 (96.85)
Yes	13,963 (3.67)	54,342 (3.14)
Missing	28 (0.01)	98 (0.01)
DM before childbirth[a]		
No	370,820 (97.55)	1,720,674 (99.45)
Yes	9,329 (2.45)	9,550 (0.55)
Smoking before childbirth[a]		
No	197,984 (52.08)	1,105,128 (63.87)
Yes	73,680 (19.38)	296,994 (17.17)
Missing	108,485 (28.54)	328,102 (18.96)
Obesity before childbirth[a]		
No	220,994 (58.13)	1,189,403 (68.74)
Yes	25,475 (6.70)	104,861 (6.06)
Missing	133,680 (35.17)	435,960 (25.20)

Abbreviations: CVD = cardiovascular disease; DM = diabetes mellitus.
[a] Information was available only for women.

analyses with autoimmune disease identified based only on primary diagnosis. Seventh, to enhance exposure specificity, we repeated the analyses by defining autoimmune diseases as confirmed by at least 2 diagnostic records for the same condition. Eighth, we defined the exposure as any autoimmune disease excluding acute, monophasic autoimmune conditions

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such as Guillain-Barre syndrome and acute disseminated encephalomyelitis. Ninth, to account for subclinical autoimmune disease, that is, the fact that some individuals are likely to have been diagnosed years after the onset of disease symptoms, we considered individuals with autoimmune disease as exposed at 1, 2, and 3 years, respectively, before their diagnosis. Finally, for analyses of specific types of autoimmune diseases involving multiple comparisons, we applied the Benjamini-Hochberg false discovery rate (FDR) correction.

All statistical analyses were conducted using R version 4.3.1.

Standard Protocol Approvals, Registrations, and Patient Consents

The study was approved by the Research Ethics Committee at Karolinska Institutet in Stockholm (Nos. 2016/288-31/1, 2021-03315, and 2025-04808-02). Individual consent is not required in case of only register-based research.

Data Availability

Qualified Swedish researchers may access similar data on approval from the regional ethical review board and the relevant registry authorities, including the Swedish National Board of Health and Welfare and Statistics Sweden.

Results

Participant Characteristics

A total of 3,824,528 individuals were included in the final analyses. Among them, 608,043 individuals had at least 1 record of autoimmune disease diagnosis during the study period (Table 1). Individuals with autoimmune disease had lower education; were more often female; and more often had CVD, DM, and a family history of CVD before baseline compared with the group with no record of autoimmune diseases. Among women, DM and smoking before childbirth were more common in the exposed group than in the unexposed group. The prevalence of most individual autoimmune diseases was low (eTable 2), with the most common being autoimmune thyroiditis (4.71%), psoriasis vulgaris (3.17%), and type 1 diabetes (1.91%).

Association Between Autoimmune Diseases and Incident SAH in the Main Analysis

A total of 10,311 individuals were diagnosed with SAH during the 83,511,956 person-years of follow-up. The overall adjusted HR for SAH in individuals with autoimmune disease was 1.18 (95% CI 1.11–1.25) (Figure 1). We observed an increased risk of incident SAH for several of the autoimmune disease categories, including endocrine autoimmune diseases, inflammatory arthritis, connective tissue autoimmune diseases, blood autoimmune diseases, nervous system autoimmune diseases, and gastrointestinal autoimmune diseases. However, for some autoimmune disease categories (vasculitis and skin autoimmune diseases) with a low prevalence, the CIs were large (Figure 1). Several specific autoimmune diseases were associated with incident SAH, including type 1 diabetes, rheumatoid arthritis, SLE, psoriasis vulgaris, multiple sclerosis, primary biliary cirrhosis, and IgA nephropathy (Figure 2). Results from the flexible parametric survival model suggested that the point estimate for the association between autoimmune disease and SAH risk was highest earlier during the follow-up and diminished over time (eFigure 1). The stratified analyses did not yield major differences in the association between autoimmune diseases and SAH risk across various subgroups (eFigures 2–6).

Sensitivity Analyses

Adjustment for additional maternal factors, that is, smoking, obesity, DM, and hypertension before childbirth among those with available data on these covariates, somewhat lowered the HRs, except for type 1 diabetes (eTables 3–6). Adjustment for tobacco and alcohol use disorders yielded similar results to those in the main analysis (eTable 7). The HRs for SAH associated with exposure to specific autoimmune diseases confirmed by at least 2 records of autoimmune diagnosis were higher than those confirmed by only 1 diagnosis record (eTable 8). The associations of SAH defined based on inpatient and specialized outpatient care and the main cause of death according to overall autoimmune disease and specific autoimmune disease categories were similar to those observed when SAH was defined by only inpatient care and main cause of death (eFigure 7).

The results were largely similar to those in the main analyses: (1) in sibling analysis, (2) when adjusting for a family history of autoimmune diseases and CVD, (3) defining exposure based on only 1 specific autoimmune disease, (4) excluding patients with comorbidities diagnosed within 3 months of their SAH diagnosis, (5) starting follow-up with a three-month lag after the autoimmune disease diagnosis, (6) defining exposure based only on primary diagnosis of autoimmune disease, and (7) excluding Guillain-Barre syndrome and acute disseminated encephalomyelitis from the exposure (eTable 9). Setting the exposure time 1, 2, or 3 years before the first record of autoimmune disease yielded somewhat stronger associations than those in the main analyses. The diagnosis of autoimmune disease before baseline was also associated with an increased SAH risk. Excluding individuals with autoimmune diseases before baseline or analyses with autoimmune disease diagnoses made after baseline—both before and after the median follow-up time—resulted in weaker associations (eTable 9). After FDR adjustment, the associations for type 1 diabetes, rheumatoid arthritis, and primary biliary cirrhosis remained statistically significant ($q < 0.05$) in the multivariable-adjusted model (eTable 10).

Discussion

This study showed that autoimmune disease was associated with an increased risk of incident SAH. The increased risk persisted after considering a wide range of potential confounders in multivariate models or unmeasured familial

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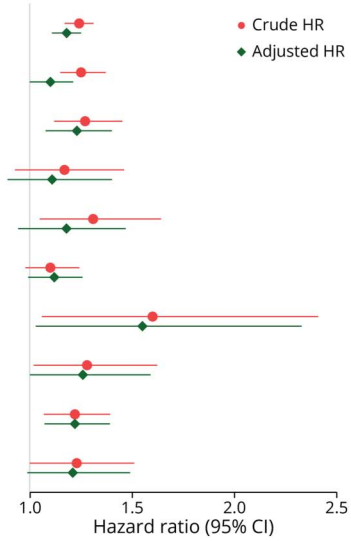
Figure 1 Crude and Adjusted Hazard Ratios and 95% CIs for Subarachnoid Hemorrhage Associated With Overall and Specific Categories of Autoimmune Diseases

Disease	Exposed events (per 10,000 PYs)	Nonexposed events (per 10,000 PYs)	Crude HR (95% CI)	Adjusted HR (95% CI)
Any autoimmune disorder	1,388 (1.77)	8,923 (1.18)	1.24 (1.17–1.31)	1.18 (1.11–1.25)
Endocrine autoimmune diseases	526 (1.86)	9,785 (1.21)	1.25 (1.15–1.37)	1.10 (1.00–1.21)
Inflammatory arthritis	230 (2.00)	10,081 (1.22)	1.27 (1.12–1.45)	1.23 (1.08–1.40)
Vasculitis	76 (1.98)	10,235 (1.23)	1.17 (0.93–1.46)	1.11 (0.89–1.40)
Connective tissue autoimmune diseases	79 (2.06)	10,232 (1.23)	1.31 (1.05–1.64)	1.18 (0.94–1.47)
Skin autoimmune diseases	272 (1.68)	10,039 (1.23)	1.10 (0.98–1.24)	1.12 (0.99–1.26)
Blood autoimmune diseases	23 (2.28)	10,288 (1.23)	1.60 (1.06–2.41)	1.55 (1.03–2.33)
Nervous system autoimmune diseases	71 (1.89)	10,240 (1.23)	1.28 (1.02–1.62)	1.26 (1.00–1.59)
Gastrointestinal autoimmune diseases	225 (1.61)	10,086 (1.23)	1.22 (1.07–1.39)	1.22 (1.07–1.39)
Other autoimmune diseases	93 (1.80)	10,218 (1.23)	1.23 (1.00–1.51)	1.21 (0.99–1.49)



The analyses were adjusted for sex, age, country of birth, household income, education, marital status, history of diabetes mellitus, and history of cardiovascular diseases. HR = hazard ratio; PY = person-year.

confounding through a family-based design. Similarly, we observed increased risks of SAH when examining different groups of autoimmune disease separately, that is, endocrine autoimmune diseases, inflammatory arthritis, connective tissue autoimmune diseases, blood autoimmune diseases, nervous system autoimmune diseases, and gastrointestinal autoimmune diseases, as well as for specific autoimmune diseases, that is, type 1 diabetes, rheumatoid arthritis, SLE, psoriasis vulgaris, multiple sclerosis, primary biliary cirrhosis, and IgA nephropathy. For some other autoimmune diseases, the confidence intervals were very wide, reflecting the limited statistical power due to the rarity of these specific autoimmune conditions and SAH.

Our finding regarding the association between autoimmune diseases and SAH risk is consistent with previous research on the role of autoimmune diseases in cardiovascular health. Some studies also reported a potential association between SLE,[16] type 1 diabetes,[26] and risk of hemorrhagic stroke while other studies reported no association.[27-29] One study found that patients with atrial fibrillation and autoimmune disease had a higher risk of hemorrhagic events compared with those without autoimmune disease.[30] These findings suggested that autoimmune activity may influence both ischemic and hemorrhagic stroke subtypes. It is important to note that some studies showed that SLE was associated with a higher risk of SAH.[16,17,19] Another study reported that there were elevated risks of SAH after hospital admission for several autoimmune diseases.[18] Evidence comprehensively examining the association between a wide range of autoimmune diseases and the risk of SAH in a large prospective cohort is limited. By examining the broad spectrum of autoimmune diseases over a 23-year follow-up and using validated clinical data with rigorous confounder adjustment, this study adds further evidence to the literature.

The association between various autoimmune diseases and an increased risk of SAH observed in this study may be underpinned by several related mechanisms. First, chronic systemic inflammation—a central feature of autoimmune diseases—can damage the vascular endothelium, increase oxidative stress, and weaken structural components of the vessel wall, making aneurysm formation and rupture more likely.[5-8] This effect may be strongest around the time of diagnosis, when disease activity and inflammation are typically highest, which could help explain the somewhat higher risk seen earlier during follow-up. Second, in some connective tissue autoimmune diseases and systemic vasculitides, direct immune-mediated inflammation of the cerebral arterial wall can lead to the formation of vasculitic aneurysms, which are prone to rupture and may underlie a subset of spontaneous SAH cases.[31-35] Third, several autoimmune diseases are associated with prothrombotic states and abnormal coagulation, which may paradoxically rise the risk of both thrombotic and hemorrhagic cerebrovascular events, especially early in the disease course.[36,37] Fourth, long-term treatment with immunosuppressive drugs such as glucocorticoids may increase vascular fragility, alter lipid metabolism, and cause secondary hypertension—all recognized risk factors of SAH.[38-40] Similarly, in type 1 diabetes, which is managed primarily with hormone replacement rather than immunomodulatory therapy, chronic hyperglycemia and metabolic disturbances may

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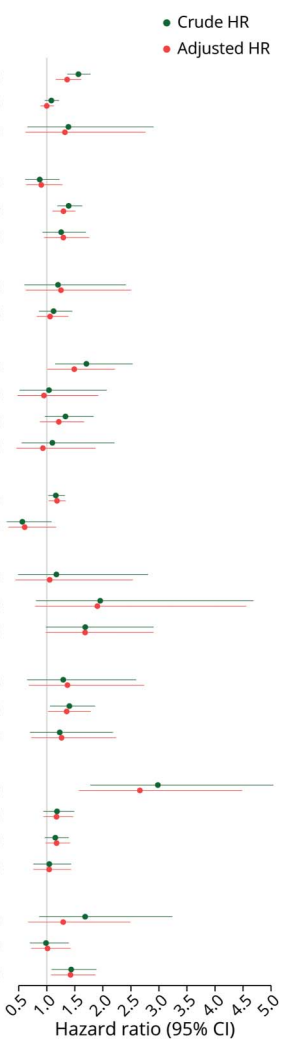
Figure 2 Crude and Adjusted Hazard Ratios and 95% CIs for Subarachnoid Hemorrhage According to Specific Autoimmune Diseases

Disease	Exposed events (per 10,000 PYs)	Nonexposed events (per 10,000 PYs)	Crude HR (95% CI)	Adjusted HR (95% CI)
Endocrine autoimmune diseases				
Type 1 diabetes	243 (2.37)	10,068 (1.22)	1.56 (1.37–1.77)	1.36 (1.15–1.61)
Autoimmune thyroiditis	302 (1.60)	10,009 (1.23)	1.08 (0.96–1.21)	1.00 (0.89–1.12)
Primary adrenocortical insufficiency	7 (2.05)	10,304 (1.23)	1.38 (0.66–2.90)	1.32 (0.63–2.76)
Inflammatory arthritis				
Reactive arthritis	33 (1.28)	10,278 (1.23)	0.87 (0.62–1.22)	0.90 (0.64–1.27)
Rheumatoid arthritis	165 (2.26)	10,146 (1.23)	1.39 (1.19–1.62)	1.29 (1.10–1.50)
Ankylosing spondylitis	41 (1.90)	10,270 (1.23)	1.25 (0.92–1.69)	1.29 (0.95–1.75)
Vasculitis				
Henoch-Schönlein purpura	8 (1.02)	10,303 (1.23)	1.20 (0.60–2.40)	1.25 (0.63–2.50)
Polymyalgia rheumatica	58 (2.22)	10,253 (1.23)	1.12 (0.86–1.45)	1.06 (0.82–1.38)
Connective tissue autoimmune diseases				
Systemic lupus erythematosus	25 (2.46)	10,286 (1.23)	1.70 (1.15–2.52)	1.49 (1.01–2.21)
Systemic sclerosis	8 (1.5)	10,303 (1.23)	1.03 (0.52–2.06)	0.95 (0.48–1.91)
Sjögren syndrome	38 (2.31)	10,273 (1.23)	1.33 (0.97–1.83)	1.21 (0.88–1.66)
Mixed connective tissue disease	8 (1.63)	10,303 (1.23)	1.10 (0.55–2.20)	0.93 (0.46–1.86)
Skin autoimmune diseases				
Psoriasis vulgaris	262 (1.78)	10,049 (1.22)	1.16 (1.03–1.31)	1.18 (1.04–1.33)
Vitiligo	9 (0.77)	10,302 (1.24)	0.56 (0.29–1.08)	0.60 (0.31–1.16)
Blood autoimmune diseases				
Pernicious anemia	55 (1.95)	10,306 (1.23)	1.17 (0.49–2.80)	1.05 (0.44–2.53)
Autoimmune hemolytic anemia	5 (3.08)	10,306 (1.23)	1.95 (0.81–4.68)	1.90 (0.79–4.56)
Idiopathic thrombocytopenic purpura	13 (2.14)	10,298 (1.23)	1.68 (0.98–2.90)	1.68 (0.98–2.90)
Nervous system autoimmune diseases				
Acute disseminated encephalomyelitis	8 (1.8)	10,303 (1.23)	1.29 (0.65–2.59)	1.37 (0.68–2.73)
Multiple sclerosis	50 (2.02)	10,261 (1.23)	1.40 (1.06–1.85)	1.35 (1.02–1.78)
Guillain-Barre syndrome	12 (1.98)	10,299 (1.23)	1.23 (0.70–2.17)	1.26 (0.72–2.23)
Gastrointestinal autoimmune diseases				
Primary biliary cirrhosis	14 (5.31)	10,297 (1.23)	2.98 (1.77–5.04)	2.66 (1.57–4.49)
Crohn disease	76 (1.60)	10,235 (1.23)	1.18 (0.94–1.48)	1.17 (0.94–1.47)
Ulcerative colitis	116 (1.60)	10,195 (1.23)	1.15 (0.96–1.38)	1.17 (0.98–1.41)
Celiac disease	39 (1.16)	10,272 (1.23)	1.04 (0.76–1.43)	1.04 (0.76–1.43)
Other autoimmune diseases				
Acute rheumatic fever and chorea	9 (2.75)	10,302 (1.23)	1.68 (0.87–3.23)	1.29 (0.67–2.49)
Sarcoidosis	34 (1.52)	10,277 (1.23)	0.98 (0.70–1.38)	1.01 (0.72–1.42)
IgA nephropathy	51 (1.95)	10,260 (1.23)	1.43 (1.09–1.88)	1.42 (1.08–1.87)



Autoimmune diseases with SAH occurring in fewer than 5 individuals/cell were not presented. The analyses were adjusted for sex, age, country of birth, household income, education, marital status, and history of diabetes mellitus and cardiovascular diseases.

contribute to long-term vascular vulnerability, potentially increasing SAH risk through mechanisms distinct from classical immune-mediated inflammation. Finally, some autoimmune diseases are linked to accelerated atherosclerosis and vascular remodeling, which can further increase cerebrovascular vulnerability.[7,41]

This study has several limitations. First, although we leveraged a large, nationwide cohort with high-quality registry data based on automatic participation of all citizens virtually without loss to follow-up, reliance on diagnostic codes may have led to misclassification because some autoimmune diseases or SAH events may be underreported or inaccurately coded. However, previous validation studies support the generally high accuracy of these registers for SAH and several autoimmune diseases.[42-45] In addition, it is possible that individuals who died of SAH before hospitalization and who did not undergo autopsy may not have been identified as SAH cases, which could introduce some misclassification in the outcome. We performed sensitivity analyses with alternative definitions for the exposures and the outcomes to address concern in our study; we observed results in these analyses that were largely similar to those in the main analyses. Second, it is possible that patients being treated for conditions related to SAH had a higher probability of having a subclinical autoimmune disease detected. Nevertheless, in our sensitivity analyses that aimed to address this concern, the observed associations remained robust. Third, despite adjustment for a broad range of confounders, residual confounding cannot be fully excluded, especially from lifestyle-related factors such as smoking, alcohol use, and physical activity, which are not captured by Swedish health registers. Smoking, in particular, represents an important confounder, given its established associations with both autoimmune diseases and risk of

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SAH.[46-49] However, reliable information on smoking status is largely unavailable in national registers; thus, we were unable to consider this confounder in the entire cohort. In a subcohort of women with available data, additional adjustment for smoking before childbirth attenuated the association between autoimmune diseases and SAH, suggesting that unmeasured smoking in the main cohort could have contributed to residual confounding. Our sibling comparison design, however, helped to account for shared familial and genetic confounders. Finally, because the cohort was restricted to the Swedish population, the findings may not be directly generalizable to other settings with different genetic, environmental, or health care characteristics. Further studies incorporating more detailed clinical and lifestyle information are warranted to deepen understanding of the link between autoimmune diseases and SAH.

In summary, this study provides additional evidence that both overall autoimmune disease and several specific subtypes are linked to an elevated risk of incident SAH. The consistent associations observed may underscore the potential value of incorporating autoimmune conditions into cerebrovascular risk profiling. If confirmed by future research, these findings could help guide SAH prevention strategies for individuals with autoimmune diseases.

Author Contributions

H. Zhang: drafting/revision of the manuscript for content, including medical writing for content. I. Janszky: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. E.V. Arkema: drafting/revision of the manuscript for content, including medical writing for content. H. Wang: drafting/revision of the manuscript for content, including medical writing for content. J. Möller: drafting/revision of the manuscript for content, including medical writing for content. Y. Liang: drafting/revision of the manuscript for content, including medical writing for content. X. Mo: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. K.D. László: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design.

Study Funding

The study was supported by grants from the Heart and Lung Foundation (20230493) and the Swedish Research Council for Health, Working Life and Welfare (2015-00837).

Disclosure

The authors report no relevant disclosures. Go to Neurology.org/N for full disclosures.

Publication History

Received by *Neurology*® August 29, 2025. Accepted in final form December 1, 2025. Submitted and externally peer reviewed. The handling editor was Editor-in-Chief José Merino, MD, MPhil, FAAN.

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